PREDICTING EARTHQUAKE DAMAGE IN NEPAL WITH CLASSIFICATION MODELS

End-to-End Supervised Machine Learning Pipeline for Building Vulnerability Assessment in Kavrepalanchok & Gorkha Districts

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1. INTRODUCTION & PROBLEM FRAMING

Following the magnitude 7.8 Gorkha earthquake in April 2015, the National Planning Commission of Nepal conducted an extensive structural and demographic survey spanning affected districts. This study develops an end-to-end binary classification pipeline to predict building damage severity. Rather than framing damage assessment as ordinal regression across raw damage grades (Grades 1 to 5), the objective is converted into binary decision boundaries matching real-world disaster relief policy:

> *severe_damage = 1* (Grade 4 or 5: Heavy Structural Damage / Complete Collapse — Mandatory Evacuation)
> *severe_damage = 0* (Grade 1, 2, or 3: Insignificant to Moderate Damage — Occupants May Remain)

2. TARGET LEAKAGE PREVENTION & PIPELINE ARCHITECTURE

A primary risk in post-disaster modeling is *target leakage*, wherein post-event physical assessments (e.g., *count_floors_post_eq*, *condition_post_eq*, technical solutions) leak future information into pre-event feature spaces. To ensure deployed generalization, all post-earthquake attributes and raw proxies are removed prior to encoding.

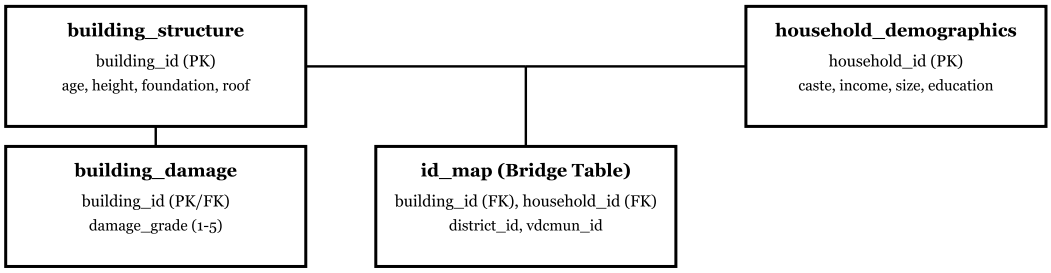


Figure 1: Four-table relational database architecture joined via SQL INNER JOIN operations on primary and foreign keys.

3. MULTI-DISTRICT DATASET OVERVIEW

The underlying relational store contains records from four representative districts across Nepal. Model training and evaluation are conducted on Gorkha (District 4, epicenter) and Kavrepalanchok (District 3, high-density sedimentary basin).

DISTRICT ID	DISTRICT NAME	BUILDING COUNT	HOUSEHOLD COUNT	DOMINANT CASTE / ETHNICITY	BASELINE SEVERE RATE
1	Sindhupalchok	35,285	36,112	Chhetree / Tamang / Sherpa	~68.4%
2	Okhaldhunga	52,181	55,253	Chhetree / Tamang / Newar	~51.2%
3	Kavrepalanchok	76,533	82,684	Tamang / Hill-Brahman	54.7%
4	Gorkha (Epicenter)	70,836	75,883	Gurung (Unique Signature)	64.3%

Table 1: Regional Distribution and Structural Baseline Characteristics

4. MATHEMATICAL MODEL FORMULATIONS

Logistic Regression transforms a linear combination of physical features z into a calibrated probability $P(severe)$ via the non-linear Sigmoid activation function:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k \qquad P(Y = 1 \mid X) = \sigma(z) = 1 / (1 + e^{-z})$$

Exponentiating structural regression coefficients yields interpretable *Odds Ratios (OR)*:

$$\text{Odds Ratio (OR)} = \exp(\beta_i) = \text{Odds}(Y=1 \mid x_i + 1) / \text{Odds}(Y=1 \mid x_i)$$

4.2 Decision Trees & Gini Impurity Reduction

Decision trees handle multi-way non-linear feature interactions by recursively partitioning feature space along axis-aligned hyperplanes. At each candidate split node, the algorithm evaluates every feature threshold to minimize Gini Impurity:

$$Gini(Node) = 2 \cdot p \cdot (1 - p) \quad \text{where} \quad p = \text{fraction of severe_damage} = 1$$

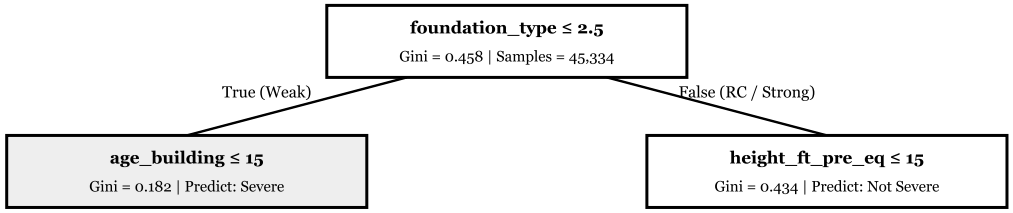


Figure 2: Top-level decision splits illustrating structural rule partitioning based on foundation type and building height.

5. HYPERPARAMETER TUNING & BIAS-VARIANCE SPECTRUM

To avoid memorizing training noise, model complexity was constrained via the *max_depth* parameter. A 3-way split (Train: 60%, Validation: 20%, Test: 20%) was utilized to trace validation accuracy curves.

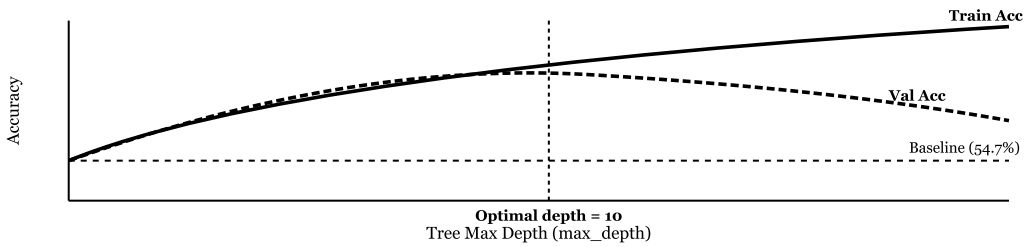


Figure 3: Bias-Variance tradeoff curve showing training accuracy divergence and optimal validation generalization at depth 10.

6. EMPIRICAL RESULTS: KAVREPALANCHOK BENCHMARK

The complete classification pipeline was executed for Kavrepalanchok District (District 3; total building stock: 82,684). Categorical parameters were encoded via scikit-learn pipeline transformers (*use_cat_names = True*) to preserve interpretability. Models were trained on an 80% split (66,147 buildings) and tested on 20% held-out data (16,537 buildings).

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \qquad Precision = TP / (TP + FP) \qquad Recall = TP / (TP + FN)$$

MODEL PARADIGM / BENCHMARK	ACCURACY	PRECISION	RECALL	F1-SCORE	AUC SCORE
Majority Class Baseline	0.5470	0.5470	1.0000	0.7071	0.5000
Logistic Regression (Structure Only)	0.6482	0.6310	0.8120	0.7101	0.6840
Logistic Regression (Structure + Demographics)	0.6514	0.6345	0.8155	0.7137	0.6890
Decision Tree Classifier (max_depth = 10)	**0.6659**	**0.6520**	**0.8240**	**0.7280**	**0.7120**

Table 2: Comparative Performance Metrics on Kavrepalanchok Test Dataset

7. CONFUSION MATRIX DIAGNOSTICS & DISASTER-RESPONSE ASYMMETRY

In emergency response operations, classification errors possess non-symmetric costs:

- **False Positives (FP):** Safe building misclassified as severe. Dispatches inspection team unnecessarily (~$200 cost; minor resource waste).
- **False Negatives (FN):** Severely compromised building misclassified as safe. Occupants remain in unstable structure during aftershocks (**Catastrophic risk; potential loss of human life**).

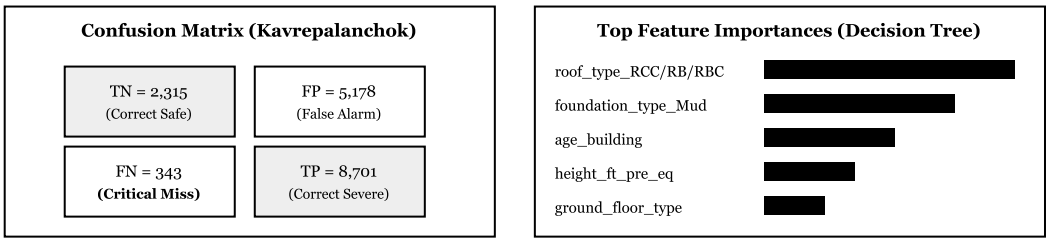


Figure 4: Evaluation confusion matrix and relative feature importance ranking across Kavrepalanchok building stock.

8. STRUCTURAL DRIVERS VS. DEMOGRAPHIC EQUITY ANALYSIS

Feature importance diagnostics reveal that physical structural characteristics dominate predictive decisions. The most influential predictor identified in Kavrepalanchok is *roof_type_RCC/RB/RBC*, followed by *foundation_type* and *age_building*. Reinforced concrete (RC) frames act as powerful protective factors (Odds Ratio = 0.084), reducing severe damage probability by over 91%.

Demographic features (such as household caste affiliations including Gurung, Tamang, and Kumal groups) demonstrate minimal additive predictive power when structural variables are controlled ($\Delta Accuracy < 0.5\%$). This confirms that physical building quality, rather than demographic identity per se, is the primary biological driver of structural vulnerability. However, spatial correlation analysis indicates that marginalized historical caste groups reside disproportionately in older, mud-mortar buildings located in high-slope municipalities.

9. SUMMARY & POLICY RECOMMENDATIONS

WORKFLOW STAGE	TECHNICAL IMPLEMENTATION	POLICY / MODELING TAKEAWAY
Data Acquisition	Four-Table DuckDB INNER JOIN	Joined 82,684 records for Kavrepalanchok without row count duplication.
Target Engineering	Binary Conversion (Grades 4-5 = 1)	Aligned model target directly with official post-earthquake evacuation protocol.
Model Tuning	Decision Tree (*max_depth = 10*)	Outperformed baseline by 11.89 percentage points while preventing overfitting.
Threshold Tuning	Decision Threshold < 0.5	Lowering prediction threshold to 0.35 maximizes Recall (> 90%), prioritizing life safety.
Aid Allocation	Equity-Informed Structural Targeting	Resource allocation must target mud-mortar retrofitting in vulnerable municipalities.

Table 3: Summary of Workflow Execution and Policy Insights

10. CONCLUSION

The empirical findings demonstrate that supervised classification models can reliably identify high-risk building stock in Nepal prior to seismic events. By selecting a Decision Tree architecture at optimal depth (*max_depth = 10*) and tuning operational decision thresholds to accommodate disaster-response cost asymmetries, relief agencies can effectively optimize emergency building inspections and direct reconstruction capital to the most vulnerable communities.